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Survey paper

A comprehensive survey on LEACH-based clustering routing protocols in Wireless Sensor Networks

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ABSTRACT

During the previous few years, Wireless Sensor Network (WSN) appears as an active research domain due to the wide use of this technology in several applications, such as military, health, automation, and so forth. WSN supervises physical attributes where human activity is difficult or impossible. The WSN is a set of multiple sensor nodes that are haphazardly deployed in a specific space. These nodes sense data and record values continuously and send these data to the Base Station (BS) through other sensor nodes. Several issues are encountered in the WSNs, including energy consumption, deployment of sensor nodes, routing algorithms, energy efficiency, Cluster-Head (CH) selection, robustness, etc. Numerous routing protocols have been developed by researchers to resolve these constraints, and several techniques of optimization are proposed to define the optimal road between the transmitter and the receiving node. Hierarchical routing protocols increase the network lifetime by enhancing its performance. The most popular hierarchical method is LEACH (Low Energy Adaptive Clustering Hierarchy) that grouped sensor nodes into clusters. Each cluster consists of normal nodes and is controlled by a CH, which is elected by cluster members to collect their data and send it to the BS. The LEACH protocol decreased the energy consumption in WSNs. In this survey, we provide a comprehensive study of LEACH descendant clustering protocols. This survey is the first study to classify LEACH-based routing protocols into CH selection, data transmission, and both CH selection and data transmission techniques. This survey is compared to other actual surveys. To evaluate these protocols, we look at many metrics, such as the CH selection method, communication method, the scalability, energy efficiency, mobility, localization of nodes, and so forth. According to these metrics, we propose a comparative analysis of these clustering routing protocols. This survey discusses also the strengths and limitations of each LEACH-variant protocol. Conclusively, the paper terminates with recommendations on future research fields in WSN.

1. Introduction

The last decades have seen a growing trend toward Wireless Sensor Network (WSN) because of the significant use of the WSNs in several domains, such as military, civilian, and industrial. WSN is a network constituted by a group of micro sensor nodes deployed randomly in a specific area, and at least one sink (or Base Station (BS)). In WSNs, there are diverse kinds of sensor nodes: source sensor node (normal node), intermediate sensor node (especially Cluster-Heads (CHs) in clustered networks), and BS [1]. These micro sensor nodes can monitor physical and environmental phenomenons like temperature, pollution, heartbeat, etc. Sensor node devices collect data and send it to the sink, then through the internet, the user can get information from the sink as shown in Fig. 1 The main objectives of WSN are: (1) control a specific zone; (2) detect and register the incidence of events in this supervised area; (3) measure the appropriated criterion [2].

In WSN, sensor nodes are alimented by batteries (in most of the cases they are neither rechargeable nor replaceable). Energy is consumed in collecting data, treatment, and transmitting or receiving packets. Therefore, the energy consumption is the major constraint in WSNs. Besides, another constraint that influences and decreases the energy efficiency of sensor nodes is the redundancy of data. Consequently, economizing energy to increase network lifetime is a challenge in WSN. In this context, many studies of WSN show the importance of routing protocols.

The routing protocol is a procedure of choosing the right route for the data to move from source to destination [3]. The objective of

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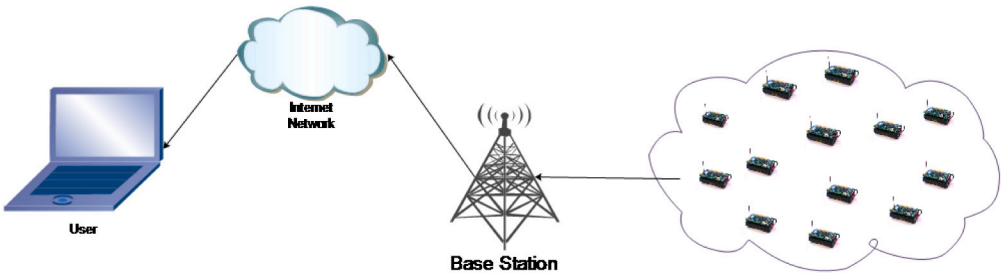


Fig. 1. WSN architecture.

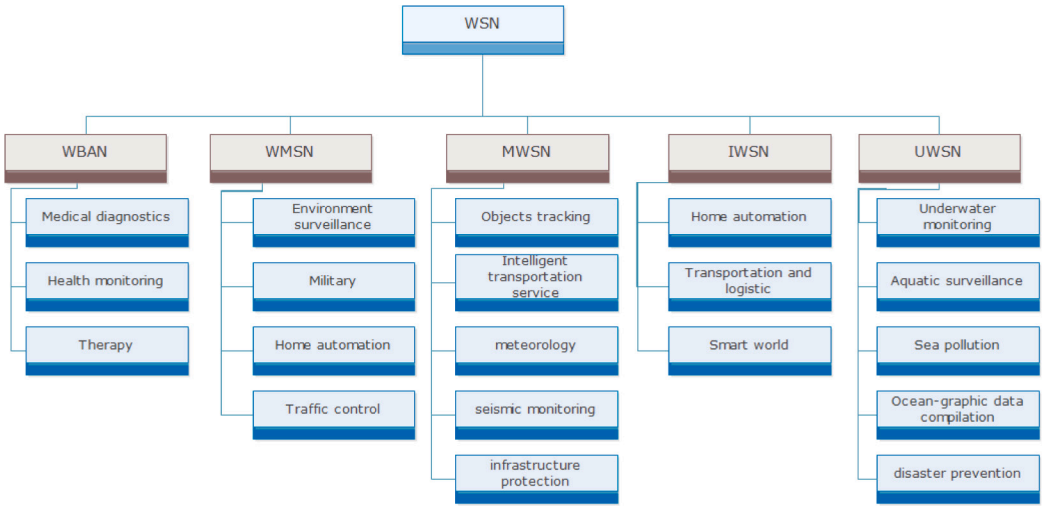


Fig. 2. WSN applications.

routing protocols is to realize the scalability of the network, strengthen data transfer, and energy efficiency [4]. During recent years, many routing protocols have been developed for the WSN to extend the network lifetime through decreasing the power consumption. These proposed routing protocols depend on several metrics, like communication model, network structure, and topology [5].

The literature review is based primarily on hierarchical routing in which the different techniques of the cluster-based routing protocol are considered [6].

Clustering puts an efficient technique for energy balancing in WSNs using data aggregation [7]. Clustering routing protocols are used in WSN to reach energy-efficient because the clustering method decreases the number of packets routed in the network. In WSNs, clustering has various advantages like more scalability, less load, less energy consumption, data aggregation, collision avoidance, load balancing, latency reduction, fault-tolerance, more robustness, a guarantee of connectivity and maximizing of the network lifetime, etc.

Less energy consumption: data aggregation serves to reduce communications data and conserve energy. Besides, intra-cluster and inter-cluster communications allow less energy consumption for the network since they decrease the cost of the distance between the farthest nodes and the sink. Moreover, the fact that just the CHs that responsible for transmitting data to the sink, it also helps to conserve energy consumption.

Data Aggregation: Serves to group the multiple data of nodes to suppress redundant transmission and furnish aggregated data to the sink [8]. Consequently in clustering, CH aggregates the received data from cluster members, then forwards the grouped data to the sink, which allows economic energy.

Less Load: In the clustering scheme, the aggregation or fusion is effected by the CHs, and clusters members communicate with their corresponding CHs, which help significantly to reduce transmission data and conserve energy. Furthermore, the length of the stored routing table on each sensor node is reduced since roads are established in clusters [9].

One of the most famous clustering routing protocols of WSN is LEACH [10]. In LEACH, sensor nodes are gathered into groups named clusters. Each cluster selects one node to become a leader (cluster-head) and the rest of the nodes become cluster members (or non-cluster heads) [11]. Every common node collects its own information from the environment and sends it to its CH. The CH groups data received from cluster members and does compression, then sends aggregated data to the sink.

The LEAH protocol ameliorates the lifetime of WSNs by decreasing the number of transmission packets via clusters formation. However, it has numerous drawbacks, such as the random selection of CHs without considering either the distance or the current energy of nodes. Another drawback is the single-hop used by CHs to communicate with the sink, which makes LEACH is not applicable for large networks.

Several surveys [12–16] have been made to examine the popular LEACH protocol and its descendant clustering hierarchical algorithms by using different classifications based on different metrics, like the clustering method, mobility of nodes, energy efficiency, and scalability.

The existing survey on the LEACH improvement protocols has been mostly restricted to limited comparisons of some criteria. Thus, some of them have not treated LEACH descendants protocols in much detail. The goal of this survey is to review recent research into the LEACH based routing protocols. The present work provides a recent survey of clustering routing protocols improved of LEACH protocol. This survey explores many LEACH variant protocols by using a novel taxonomy, which relies upon on kinds of metrics, such as energy-efficiency, clustering method, and communication technique.

In clustering hierarchically based protocols, both CH selection and data transmission techniques have importance in extending the life-cycle of WSN. For this reason, this survey proposes a new taxonomy,

which classified LEACH enhancement protocols into three classes: (1) clustering protocols based on CH selection to improve the LEACH protocol; (2) protocols based on data transmission techniques to enhance LEACH; (3) protocols based on both CH selection and data transmission methods. Then in each class, protocols are classified in terms of some criteria, such as the residual energy, mobility, distance, single-hop, multi-hop technique, and so forth.

The first-class concerns LEACH descendant protocols based on the CH selection. This class is established in terms of some metrics, like residual energy, energy-efficiency, mobility of nodes, distance, and centralization. The second one concerns protocols based on data transmission techniques, which groups protocols into three categories: protocols that use the single hop to route their data, protocols use the multi-hop, and protocols that use both techniques. The third class contains protocols that have improved both CH selection and data transmission compared to the classical LEACH algorithm.

The contributions of the recent survey are as follows:

- LEACH descendant clustering routing protocols are classified based on CH selection and data transmission techniques through a new taxonomy based on numerous criteria, such as energy-efficiency, distance, residual energy, single hop, and multi-hop.
- An interesting comparison between discussed protocols based on several metrics, like energy efficiency, communication techniques, clustering methods, the CH selection, deployment, the scalability, etc. is performed to facilitate the selection of the most effective protocol for researchers based on their needs.
- A comparison of the present survey with other previous surveys.

The resting part of the survey proceeds as follows: the second section offers an overview of WSN which introduces a background of WSN, its applications, its components, and its constraints. Further, This section indicates an overview of the conventional LEACH protocol and an overview of several existing surveys. The third section introduces the proposed taxonomy of LEACH descendant protocols, which describes the routing protocols for each category and provides a comparative analysis of different discussed protocols. Future works and research challenges are given in the fourth section. Finally, the final section provides a conclusion of this survey.

2. Overview of WSN

WSN offers new technology with convincing benefits in contrast to traditional wired networks, which has the advantage of decreasing installation and servicing charges of monitoring systems [17].

WSN comprises a group of small, intelligent electronics devices recognized as sensor nodes (SNs) deployed over the network, and at least a BS connected to these sensor nodes via wireless channels. The number of SNs can vary from a few to various hundred relying on the applications [18]. Sensor nodes sense and supervise the physical and environmental phenomenon, such as health monitoring, temperature, humidity, etc. To reach the surveillance of a particular environment, sensor nodes can communicate and collaborate with others to send their gathered data to the network via the BS [19]. Fig. 1 presents the architecture of WSN.

2.1. WSN applications

Recently, WSN has been increasingly important in the world by its application in numerous domains, where human activities are very difficult [20], such as military monitoring, environment, industrial, home automation, medical surveillance, underwater, etc. We grouped these WSN's applications in taxonomy as shown in Fig. 2.

When sensor networks are used in the medical and health domain, it concerns WBAN (Wireless Body Area Network). WSN is called WMSN (Wireless Multimedia Sensor Network) when sensor nodes are used to transfer multimedia, such as images, audio, and videos. When some applications require the mobility of nodes like objects tracking, intelligent transportations service, meteorology, seismic monitoring, infrastructure protection, they called MWSN (Mobile Wireless Sensor Network). Furthermore, when sensor nodes distributed in industrial societies or in factories to control machines, it affected IWSN (Industrial Wireless Sensor Network). Besides, UWSNs (Underwater Wireless Sensor Network) distribute sensor nodes underwater to monitor sea pollution, underwater navigation, catastrophe prevention, etc.[21]

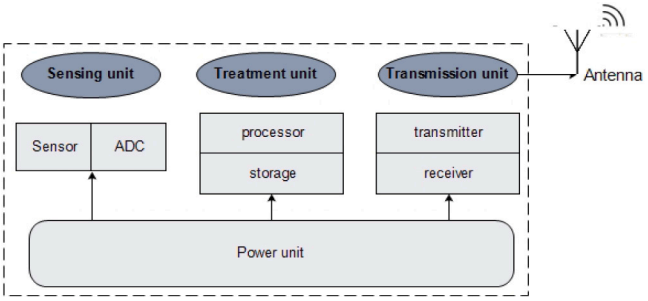


Fig. 3. WSN Units.

2.2. WSN units

WSN is a non-infrastructure network that consists of a set of autonomous SNs for detecting and monitoring an area in the environment. These SNs congregate to built a wireless network, which is deployed randomly in the desired geographical region to monitor. Every SN detects certain data and transmits it to the BS through other SNs, using wireless communication.

Every node in the WSN has four main units as illustrated in Fig. 3: (1) sensing unit that including one or more sensors for data acquisition from the environment [22], then an ADC (Analogue to Digital Converter) to convert the analog signal produced by the sensors based on the observed phenomenon to numerical signal; (2) treatment unit consists of a processor, storage, and routing protocol. This unit executes the communications protocols which allow communication between nodes; (3) wireless transmission unit involves two radio frequency modules of emission and reception that offer a wireless communication to connect the node to the network; (4) power unit (generally are batteries) is the unique source of energy that supplies all the others units.

2.3. WSN constraints

WSNs have numerous characteristics and they have imported several benefits over the traditional wired networks. In contrast, many constraints are faced by WSNs. In the first one, in almost WSN applications, the deployment of SNs is done randomly. Another constraint in WSNs is the energy, due to the limited batteries power which they are very difficult to recharge or replace them. Besides, data redundancy constraints can also decrease the energy-efficiency. WSNs have also limitations on processing capability, memory capability, communication bandwidth, and transmitting the collected data from SNs to the BS.

In WSN, the power unit provides energy to all SN components. Furthermore, SNs waste energy in data collecting, in the treatment, the transfer, or reception of packets. Therefore, the SN can dissipate its initial energy early in a short time, which influences the lifetime of the entire network. Consequently, limited power is a considerable constraint in WSN. In this context, several routing protocols are appeared to solve such scenarios.

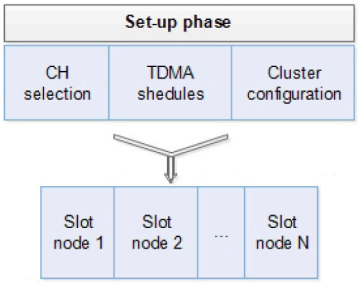


Fig. 4. Set-up stage of LEACH.

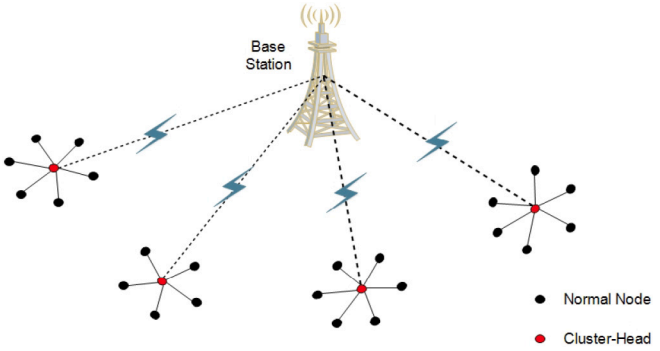


Fig. 5. LEACH architecture.

2.4. Overview of the LEACH approach

To resolve the constraints posed by WSN, several routing protocols are introduced, which serve to improve the consumption of energy-efficiency and extend the network lifetime. The most interesting protocols are the hierarchical ones, like the LEACH protocol. These protocols provide the clustering concept. In this section, firstly, we provide an overview of the LEACH protocol, and then in the second subsection, many existing surveys are discussed.

LEACH [10] was proposed to guarantee a balanced energy utilization and to enhance the efficiency of WSNs by partitioning the network into multiple clusters, and through a random CHs rotation [23]. LEACH is a MAC protocol based on the TDMA method, which is considered as the most famous and the first hierarchical routing protocol used clustering. LEACH is a routing protocol based on the clustering mechanism in which nodes are associated together to establish separate clusters. Thus, every cluster has a head node called CH that aggregates cluster members data, then sends it to the sink [24].

The tasks of the LEACH protocol are characterized by turns, and each turn is involved in two main stages, which are the set-up and the data transfer stage. The set-up stage includes the election of CHs, the formation of clusters, and the creation of the TDMA schedule as illustrated in Fig. 4.

2.4.1. Set-up stage

The first stage is intended to select CHs and form clusters in the network. Every sensor node votes to turn into a leader (or CH) of its cluster or not for the present epoch. This decision depends on the random number choosing by the sensor node at the starting of each epoch. The choosing number is between 0 and 1. The sensor node becomes a CH in the present epoch if this number is lower than the threshold function defined in Eq. (1)[10]. On the other hand, it becomes a normal node [25]. The selection of CHs following this manner leads to distribute energy between all the network nodes.

$$T(n) = \begin{cases} \frac{P_L}{1-P_L*(r \bmod (1/P_L))}, n \in C \\ 0 \qquad\qquad\qquad \text{otherwise} \end{cases} \tag{1}$$

Where P_L introduces the percentages of CHs in each epoch, r is the present epoch, and C is a set of sensor nodes that have not yet been CH in the period $1/P_L$ epochs.

Once CHs are chosen, they inform the network nodes of their status by broadcasting an advertisement message. Afterward, every sensor node joins its corresponding CH depending on the strength of the received signal from CHs. The acquired signal has high force if the CH is very near to a sensor node. Once a node chooses its join CH, this CH is informed by a request message to adding this node to its cluster members.

When clusters are established, each CH sends to its members a TDMA schedule. The TDMA schedule allows nodes to recognize their time slots for sending their collected data to CH.

2.4.2. Data communication stage

The data communication stage is composed of numerous frames and it is much longer than the first stage [26]. In this stage, each normal node sends its collected data to its potential CH using a time slot in the TDMA schedule. After the fusion of data by CHs, these data will be forwarded to the sink [27] using the CDMA code to avoid the collision.

Fig. 5 illustrated a WSN protocol based on LEACH architecture, which is split into clusters.

To simulate the LEACH protocol, an algorithm is given in [28].

2.4.3. Advantages and disadvantages of LEACH

To avoid interference between nodes, LEACH uses the TDMA schedule. To prevent collision among CHs, the CDMA code is taken up in the LEACH approach. Besides, in LEACH, nodes remain in sleep mode until their turn to send data arrives. Thus, it decreases the number of transmission packets through forming clusters and selects CHs. Although, these advantages, LEACH stills limited by several disadvantages. The deployment of CHs and the number of CHs in each round cannot be guaranteed owing to the random selection of CHs. Thereby, in LEACH, nodes with low remaining energy have the same percentage to become CH as well as nodes with high residual energy. Consequently, if these nodes become CHs, they will exhaust their energy early than other nodes. Another problem of LEACH is the CHs selection ignores the remaining energy of nodes. Despite that, the CH supports higher traffic charges than normal devices due to its role which is represented in acquiring data from all its cluster members, also aggregation and sending to the sink [29]. Then, CHs use the single-hop to direct data to the BS that makes LEACH not adopted for a large network.

2.5. Existing surveys

Authors in [12] discussed six variants of the original LEACH algorithm, which get over the non-uniform CH distribution posed in the LEACH algorithm. This survey described some LEACH-based hierarchical routing protocols, such as LEACH-C, MM-LEACH, TL-LEACH, SEP, V-LEACH, and MOD-LEACH then offered differences of each protocol from the conventional LEACH protocol, based on the energy dissipation. The drawback of this survey is that the authors did not use any classification to examine these protocols. Thus, they just considered six LEACH descendants.

The survey in [13] studied the evolution of energy consumption of several protocols derived from LEACH. This review focused on the CH election mechanism. To evaluate the performance of these descendant protocols, authors simulated and compared them to the LEACH protocol according to their energy consumption. In [13] a comparison study of various hierarchical routing protocols of LEACH is done based on certain metrics like the number of CHs and the hop count. The limitation of this survey is that no taxonomy is carried out to classify these routing protocols. Thus, merely eight variants of LEACH are

presented. The authors are concentrated on their comparison of a few factors. However, other main metrics are missing.

Mahmuddin et al. in [14] have presented a survey that described LEACH and some of its variant protocols. This study highlighted the different communication techniques between the CHs and the final destination (BS) named: single-hop and multi-hop. The authors mentioned the advantages and drawbacks of LEACH and its variant protocols based on their functionalities and their results. Similar to the previously mentioned surveys, This survey also has limitations such as the review describes ten LEACH-based protocols without offering any classification. Thus, authors did not use any metric of WSNs to compare protocols, as residual energy, scalability, energy efficiency, mobility, etc.

Authors in [15] provided a review that depicts the famous LEACH protocol and its enhancement protocols. In this survey, protocols are compared to each other based on the data aggregation, nodes mobility, and the type of scalability. This review has several disadvantages among them is that the authors talked about twelve protocols derived from LEACH without classifying them in taxonomy. Thereby, they have not highlighted other factors as well as they did not indicate the advantages and disadvantages of each discussed protocol.

In [16], Al-Shalabi et al. furnished a survey that presents an overview of LEACH and its variants. They provided also an overview of some existing reviews then they introduce their limitations. This survey put the light on CH election and clusters formation (or authors group LEACH-based routing protocols into two groups: CHs selection and clusters formation). Therefore, authors have compared some LEACH-based routing protocols based on the techniques of clusters formation and CHs selection. Thus, the advantages and weaknesses of these protocols are provided as well as their contributions. Thereby, this survey has compared these protocols depend upon certain aspects such as the residual energy, location of nodes, complexity. However, as the previous reviews, this survey remains to be developed since many metrics of WSNs did not showcase, also some protocols are not very detailed.

The previous surveys mentioned above concentrated either on CH selection criteria or cluster formation techniques or only talked over the main contributions, advantages, and disadvantages of LEACH-based protocols. On the contrary, this survey presents a detailed description of the LEACH approach and an overview of precedent surveys. This review suggests a complete classification of LEACH and its variant protocols, which focal point on criteria used on set-up phase (CHs selection and clusters formation) and criteria used on the data transmission phase.

3. LEACH descendant protocols

In this review, we have proposed a new taxonomy of LEACH descendant protocols, which focused on CH selection and data transmission techniques to classify variant protocols of LEACH. Fig. 6 shows the taxonomy of LEACH-based routing protocols.

The above taxonomy provides the classification of protocols based on CH selection and data transmission. Each class is divided into categories according to the metrics and techniques used.

3.1. CH selection-based protocols

CH selection is one of the important phases for extending the WSNs lifetime. The election of CH influences the energy-efficiency of the WSNs. All routing protocols based on CH selection have the same principle of picking the most efficient node in the network to be a CH (a leader of a cluster). Each protocol selects CH considering one or more specific parameters. Based on these parameters, these routing protocols are classified into five categories: (1) residual energy, (2) Centralization, (3) the mobility, (4) energy-efficiency, and (5) the distance. These categories are defined in the next parties.

3.1.1. Residual energy

In CH selection, residual energy is an important metric, which impacts on the network performance and the network duration. The network lifespan will be extended when the selected CH is a sensor node with the highest energy among all cluster members.

Tong et al. [30] introduced an improved version of LEACH called LEACH-B (LEACH-Balanced), where the number of CHs is guarded near-optimal. The LEACH-B is a decentralized approach used for cluster formation [31], by enhancing election CHs to make the number of CHs optimal given the remaining energy of nodes in the set-up stage. The LEACH-B protocol has three main stages: CH election, clusters formation, and data communication.

In the starting, for the first selection of CH, LEACH-B remains like the LEACH approach. After that, LEACH-B adds a second phase that considers the remaining energy. LEACH-B guarantees a uniform number of CH in each round, which is calculated through Eq. (2)[30] below, which is estimated between 3% and 5% for the total node number of 100.

$$No.of\,CHs = N * P \tag{2}$$

Where P presents the proportion of CHs in the network, and N shows the global nodes number in the entire network.

The nodes cognize just its own position and the location of the BS but do not know the position of other nodes [32]. If CHs number in the network is inferior to N*P, then a normal node with enough energy will be a CH. On the other hand, some low energy CHs will be removed. Unlike the LEACH algorithm, LEACH-B guarantees a uniform number of CHs [33]. Simulation results of the LEACH-B algorithm in [30] demonstrate that LEACH-B imported an enhancement of LEACH, which extends the network lifespan. A flowchart of the LEACH-B is given in [30].

The improved LEACH (I-LEACH) is a clustering routing protocol, which in terms of residual energy, nodes location, and neighbor number it picks out CHs [34]. In the I-LEACH approach considers the optimal cluster number in the network K_{opt} described in Eq. (3) [34].

As with several clustering routing protocols, the I-LEACH algorithm has three phases namely, CH selection, cluster formation, and data transmission. In the original LEACH algorithm, the CH selection depends only on the pre-set threshold function without considering other factors, what influences on the life span of the network. To solve this problem, the I-LEACH approach updates the threshold function introduced in LEACH by putting the residual energy, number of neighbor nodes, and the distance to the BS factors, as defined in Eq. (3) [34]. In the CH selection stage, every node produces a number between 0 and 1 randomly. If this number is less than the threshold function in Eq. (3) [34], then it is elected as CH in the present round.

$$K_{opt} = \frac{\sqrt{N}}{\sqrt{2\pi}} * \sqrt{\frac{\epsilon_{fs}}{\epsilon_{amp}}} * \frac{M}{d^2_{toBS}} \tag{3}$$

Once CHs are picking out, every sensor node joined a cluster depending upon the received signal from CHs. Nodes which are near to each other collect similar input and have a high risk of the overlap [34]. To attack this issue, in the data transfer stage of the I-LEACH protocol, each CH incorporates its data t=received from its cluster members to decrease significantly the charge of the transmitting additional data.

Table 1 introduces the advantages and disadvantages of protocols based on residual energy to elect CH.

3.1.2. Centralization

Centralized formation methods are appropriate for networks in which the treatment is central control. At each round, nodes send some information to the BS, such as the location information and the residual energy. Based on this information, the sink (BS) makes use of a particular technique to pick the CH and distributes all nodes to these clusters. A centralized technique guarantees better cluster segmentation [35].

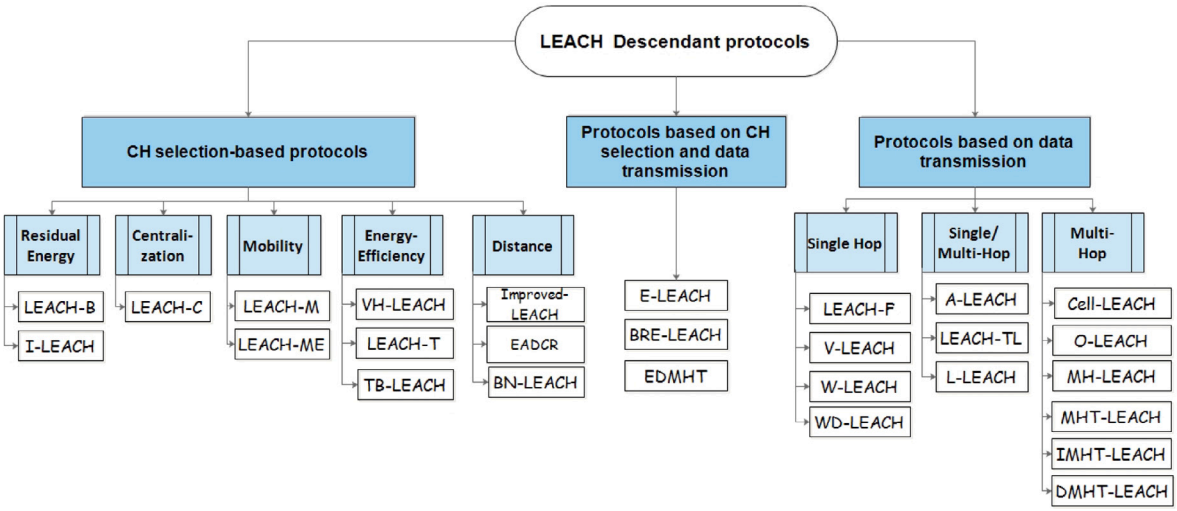


Fig. 6. Taxonomy of derived LEACH protocols.

LEACH-C is a centralized clustering algorithm implemented at the BS [36], which selects CHs according to the residual energy of nodes, then forms clusters in the network. LEACH-C uses the centralized method to choose the best CH from nominee CHs, which can effectively solve the efficiency problem posed by the classical LEACH protocol [37]. In the first phase, sensor nodes forward information about their remaining energy and location to the sink. Based on this information, the sink computes the average energy of nodes. Then, it decides which nodes are going to be CHs. Hence, nodes that have remaining energy more than the average will be selected as CHs in the present round. On the other hand, the rest becomes a normal node. The steady phase is identical to that of LEACH.

The centralization method offers popularly a better distribution and selection of CHs. Although, the number of communication between nodes and the sink decreases the life-cycle of the network. Like LEACH, the LEACH-C algorithm sends data directly to the sink in a single hop regardless of the distance, which may reduce CH lifetime [38]. Thus, in the LEACH approach, the number of CHs changes from turn to others, while in the LEACH-C approach the number of CHs is predetermined for all turns.

Table 1 introduces the advantages and disadvantages of the LEACH-C protocol.

3.1.3. Mobility

Mobility is one of the fundamental features of WSN. Mobility in wireless sensor networks can be classified taking into account: (1) physical aspects of mobility, which concerns the mobile element and the type of movement; (2) architectural aspects, such as the entity that manages the mobility process [39]. In the CH selection, mobility of nodes (or mobility of the sink) can help to choose the most efficient node as CH and improve the network lifetime.

LEACH-M (Mobile LEACH) [40] comprises the mobility factor at the conventional LEACH algorithm. This mobility concerns both CHs and cluster member nodes over the set-up phase as well as the steady phase, whereas the BS is examined as stationary as in the original LEACH. Nodes are selected as CHs based on the lowest attenuation and the minimum mobility. Then, the elected CHs diffuse their position to all nodes in their transmission scale [32].

The authors of LEACH-M considers that all nodes are initially homogeneous, and the GPS is used to locate sensor nodes. The threshold function of LEACH is used also in LEACH-M to elect CHs, although LEACH-M considers mobility of nodes along the data transmission stage contrary to LEACH.

The mobility method can face a major problem that mobile cluster members may quit its cluster when it is transferring information to its CH belongs. To solve this issue, the LEACH-M algorithm can assure that nodes remain in communication with CH or not through the TDMA schedule. At the start of each round, CH justifies the absence of any node by a message request. If the node cannot answer for two successive messages, then it is out of the CH coverage [33].

In the data transmission phase, can be faced with another issue that if any node or CH is displaced from its cluster, then the data will be lost. To resolve this problem, the LEACH-M approach uses the handover mechanism, which allows nodes to commutate to new CH if it is changed the cluster [31].

Authors in [41] proposed an improvement LEACH-M protocol refer to LEACH-ME (Mobile Enhanced-LEACH), which serves to ameliorate LEACH-M. It picks out nodes with minimum mobility comparatively to its neighbors as CHs. Each node includes CH transitions it has made all through the steady stage whilst transmitting data. Nodes transmit several transitions to its CH over the TDMA slot [42]. The main aims of LEACH-ME is a guarantee that the mobility of CHs is the most minimal relative to other nodes in the network, result that when the clusters move and change places, the disturbance due to the movement of CHs will be minimal as much as possible.

Table 1 introduces the advantages and disadvantages of protocols based on the mobility of nodes and CHs.

To simulate the LEACH-mobile and LEACH-C algorithm, algorithms are provided in [43].

3.1.4. Energy-efficiency

Energy-efficiency presents a principal challenge in routing protocols to choose CHs. When CHs are picked out with the highest energy-efficiency, the network lifespan is prolonged. There is a significant difference between the residual energy and the energy-efficiency. Therefore, routing protocols based on the residual energy to select CH, they do not necessarily consider the communication charge, the consumption of energy, etc.

Several clustering routing protocols are based on the energy-efficiency to select CH, such as VH-LEACH, LEACH-T, TB-LEACH, which are discussed in the following paragraphs.

In [44], the authors proposed an enhanced V-LEACH protocol called VH-LEACH. The principal purpose of this protocol is to change the election of CH. In the VH-LEACH algorithm, CH is selected depending on the maximum remaining energy [45].

Clusters are established and CHs are chosen. Every CH obtains the information about the remaining energy of cluster members. Afterward, the CH picks out a VH (Vice cluster-Head) which has the maximum remaining energy within the cluster [44].

The VH node stays in sleep mode until the residual energy of the main CH reaches a threshold. If the CH residual energy achieves this threshold, the VH gets up to play the CH role and picks out its VH.

Authors in [46] introduced a novel LEACH based clustering routing protocol named LEACH-T, which allows fixing the number of time slots in TDMA of each CH depending on the cluster member nodes number. In the cluster formation phase, the LEACH-T takes into account the remaining energy of nodes, thus determined an optimal number of CHs using Eq. (4) [46].

$$K_{opt} = \frac{\sqrt{N}}{\sqrt{2\pi}} * \sqrt{\frac{\epsilon_{fs}}{\epsilon_{amp}}} * \frac{M}{d^2_{toBS}} \tag{4}$$

In the steady phase, the T-LEACH involves variable time slots for distinct CHs [46]. Instead of a fixed allocation of time slots to each CH, the LEACH-T algorithm uses a dynamic technique to allocate time slots in the TDMA schedule according to the number of cluster members. That means, if the quantity of nodes is little, then only a few time slots are allotted to the CH, while if the cluster member nodes is bigger, so giant time slots should be assigned [13]. Thereby, to transfer data from CHs to the sink, it uses either single-hop or multi-hop.

TB-LEACH (Time-based Cluster-Head Selection Algorithm for LEACH) is defined in [47], which modifies the CH selection of the LEACH approach to maximize the life-cycle of the network. TB-LEACH includes two main phases: Set-up and steady stages. In the first phase, instead of the random selection of CHs in LEACH, the TB-LEACH algorithm selects CHs to depend upon a random timer. Unlike LEACH, the number of CHs in each turn is fixed and represented by a counter. At the start of each epoch, a random timer is generated by every node. If the random timer is expired and the counter does not achieve the number of CHs, the node diffuses a CH-ADV advertisement message to announce at all nodes that it becomes a CH in the current turn using CSMA [19]. Even if, the node will be a non-CH node (normal node). After that CHs are determined, the rest of the tasks are similar to the LEACH algorithm.

Generally, The TB-LEACH algorithm improves LEACH protocol by fixing the CHs number and selecting CHs depending on a random timer. Simulation results in [47] showed that the TB-LEACH enhances the network life span by up to 30% compared to the conventional LEACH protocol.

Table 1 introduces the advantages and disadvantages of protocols base on energy-efficiency to improve the set-up phase.

3.1.5. Distance

Distance between CH and the BS also has a vital impact on energy consumption. When the CH is very far from the BS, it results in a major issue in energy consumption. The CH sends aggregated data to the BS, so the distance from the CH to the BS will influence the data transmission process. Therefore, the distance factor is very important in selecting CH.

Numerous clustering routing protocols are based on the distance to pick out CHs, such as Improved-LEACH, EADCR, and BN-LEACH.

Improved-LEACH is one of the clustering routing protocols based on the distance to select CHs in the network. Authors in [48] proposed this modified LEACH protocol by considering two parameters when selecting CH: the residual energy of nodes and the distance to the BS.

Authors use Eqs. (5) and (6) [48] to calculate the residual energy of each node and the distance from CH to the BS, respectively.

$$SCRFN_1 = \frac{E_{current}}{E_{max}} \tag{5}$$

Here, E_{max} presents the initial energy of the node, while $E_{current}$ expresses the residual energy of each node.

$$SCRFN_2 = \frac{d_{bs}}{d_{far}} \tag{6}$$

Here, d_{bs} parameter denotes the distance between a node and the BS, when the distance from the farthest node in a cluster to the BS is expressed by d_{far}.

To extend the network lifetime and the scalability, functions described in Eqs. (5) and (6) are incorporated and multiplied by the probability function. The combination function is given in Eq. (7).

$$SCRFN = w_1(\frac{E_{current}}{E_{max}}) + w_2[\frac{d_{bs}}{d_{far}}] * [\begin{cases} \frac{P}{1-P*(r \bmod (1/P))}, n \in G \\ 0 \qquad\qquad\qquad \text{otherwise} \end{cases}] \tag{7}$$

Here, w_1+w_2 =1, with w_1 is the weight that is given according to the preference for energy, and w_2 is the weight that is given according to the preference for distance.

EADCR (Energy Aware Distance-based Cluster Head selection and Routing) is proposed by [49] to prolong the network lifetime of WSNs based on the FCM (Fuzzy C-Means), the residual energy of nodes, the Euclidean distance from the BS, and cluster centroid. In the EADCR, clusters are composed through making use of the FCM approach at the BS. Then, each cluster selects its CH using a fitness function. This function depends upon the Euclidean distance between the node and the BS, the distance from the node to its CH, the initial energy, and the residual energy of nodes, as defined in Eq. (8) [49].

$$F_{CH} = \begin{cases} \alpha x_1 + (1-\alpha)x_2, & (E_{res} >= E_{th}) \ \ \& \ \ (d_{n2BS} > d_{n2C}) \\ x_1, & (E_{res} < E_{th}) \ \ \| \ \ (d_{n2BS} = d_{n2C}) \\ \alpha x_1 + (1-\alpha)x'_2, & (E_{res} >= E_{th}) \ \ \& \ \ (d_{n2BS} < d_{n2C}) \end{cases} \tag{8}$$

Here, E_{res} represents the residual energy of each node, while E_{th} expresses the threshold energy described in Eq. (9). The Euclidean distance from the node n to the BS; and its centroid are presented by d_{n2BS}, d_{n2C}, respectively. x_1 represents the ratio of the remaining energy to the commencing energy of the sensor node. x_2 and x'_2 are expressed in Eq. (10). Finally, the authors are chosen the value of α equal to 0.5.

$$E_{th} = l(E_{elec} + E_{DA})n_c \tag{9}$$

Where, l introduces the number of bits, E_{elec} describes the energy consumed in transmitting or receiving 1-bit data. The energy consumption during the data aggregation process is expressed by E_{DA}. Finally, the n_c parameter is the number of cluster members.

$$x_2 = |(d_{n2BS} - d_{n2C})/d_{n2BS}|, \quad x'_2 = |(d_{n2C} - d_{n2BS})/d_{n2C}| \tag{10}$$

The CHs selection is simultaneously from cluster members. Each cluster elects the node which has the highest fitness value as a CH. The authors have implemented the rotational-based clustering concept used in the LEACH protocol. In the EADCR protocol, according to the residual energy and the Euclidean distance, a new approach for packet routing is proposed.

BN-LEACH (Bayesian Network -LEACH) [50], the authors used the model of Bayesian Network (BN) to pick out CHs in the network. In this approach, three main factors are considered, the density of nodes, the distance to the BS, and the residual energy. The BN-LEACH operations are relatively the same as the conventional LEACH approach, except that the CH selection reposes on the BN model. This model computes the probability to become CH for every node. The BN-LEACH routing protocol serves to prolong the network lifetime by extending the first node death and distributing the energy load among the sensor nodes. To distribute CHs uniformly in the network, the convenient CHs are chosen under a dynamic zoning technique and a greedy method to distribute cluster-heads uniformly. The area is divided into three regions, far, central, and near. The central area has the highest prior probability, near having medium and far has lowest [51].

The discussed improved LEACH hierarchical routing protocols based on the CH selection to ameliorate the LEACH approach are summarized in Table 2 according to the research domain, network type, modeling parameters of each protocol, and the complexity of the algorithm.

Table 1
Advantages and disadvantages of LEACH-based protocols (class 1).

Protocol	Strengths	Limitations
LEACH	Improves the life-cycle of the network through using the TDMA schedule. Equilibrates the energy consumption. Decreases the no. of transmission packets. Diminishes energy dissipation of nodes.	Assumes non-uniform deployment of CH nodes. Random election of CH. The quantity of nodes in every cluster is not evenly distributed. Uses the single hop.
LEACH-B	Uniform No. of CHs. Considers remaining energy after the first turn. Enhances the network lifespan.	Growth in overhead cost.
I-LEACH	Elects CH based on the remaining energy and nodes location.	CH integrates collected data to reduce the cost of supplementary data transmission, which is not practice for nodes that receive different data.
LEACH-C	Reaches a high no. of turns in a petty network zone. CH selection according to residual energy. The centralization method offers a better CHs distribution. Introduces an optimal no. of clusters.	Needs location information of nodes. Centralization introduces an overhead on the sink. Transmission using single hop produces an extra overhead.
LEACH-M	Mobility of both CHs and non-CH node. A very high energy-efficiency.	Supplementary overhead.
LEACH-ME	Supports the mobility of nodes. Selects CHs depend upon the minimum mobility.	Increased overhead.
VH-LEACH	CHs election on the basis of the remaining energy. Replaces CH by VH node when the main CH achieves a threshold.	Supplementary treatment for VH node. Uses the single hop to direct data from CHs to the BS.
LEACH-T	Fixes the no. of time slots in TDMA based on the no. cluster members. Offers an optimal no. of clusters.	CHs are selected randomly. Data transmission from CHs to the sink using a single hop.
TB-LEACH	Increases the life-cycle of the network. The no. of CHs is fixed.	CHs selection based on a random timer.
Improved-LEACH	Amelioration of the network performance by a well selection of CH.	Complexity of calculation.

Table 2
General characteristics of LEACH-based protocols (Class 1).

Protocol	Year	Research domain	Network type	Modeling Parameters	Complexity
LEACH	2000	Load balancing	homogeneous	Energy saving	Low
LEACH-B	2010	balancing the number of CHs	homogeneous	Remaining energy	Yes
LEACH-C	2002	Cluster head changing	homogeneous	Remaining energy and nodes position from the sink	Yes
LEACH-M	2006	Mobility of nodes	homogeneous	Mobility factor	Yes
LEACH-ME	2008	Mobility and status of nodes	homogeneous	Mobility factor	Yes
I-LEACH	2013	Energy-Efficient	homogeneous	Remaining energy, Neighbor number and nodesś location	No
VH-LEACH	2015	Multiple CHs in the cluster	homogeneous	Remaining energy	Yes
LEACH-T	2014	Optimal Number of CHs	homogeneous	Remaining energy	Yes
TB-LEACH	2008	Cluster head changing	homogeneous	Random timer	Yes
Improved-LEACH	2016	Energy-efficiency	homogeneous	Distance from node to BS	Yes
EADCR	2020	Energy-efficiency	homogeneous	Euclidean distance and Residual energy	Yes
BN-LEACH	2014	Load balancing	homogeneous	Density, distance and residual energy	No

3.2. Data transmission-based protocols

Data transmission is the second significant phase for prolonging the WSNs lifetime. This phase represents data communication between normal nodes and their CH, and between CHs and the BS. So, it largely affects the energy consumption in the network. The principal idea in all data transmission-based protocols is to route data either in a single hop or in multi-hop or both techniques. Consequently, these techniques classified these routing protocols into three categories, called, single-hop, multi-hop, both single and multi-hop, which discussed in the next parties.

3.2.1. Single hop

In this category, protocols are based on the single hop technique to route data from CHs to the BS.

Authors in [52] introduced the LEACH-F (Fixed number of cluster-LEACH) protocol, which forms clusters and selects CH using a centralized approach like the LEACH-C algorithm.

Once clusters are established, they are fixed and there is no re-clustering stage again at the starting of each epoch [53]. Therefore, clusters are fixed and just turnaround the role of CH in their cluster. Furthermore, the steady phase stays unchangeable compared to the LEACH approach. LEACH-F reassures the reduction of the set-up phase by fixing the clusters at the beginning of the phase. However, LEACH-F has many disadvantages, among them:

- Fixing clusters may oblige nodes to remain in a cluster even if the CH is very further.
- No flexibility to remove or add nodes to the network after the clusters are provided.
- Nodes cannot regulate their conduct at the death of nodes.
- The mobility of nodes cannot be managed.

The CH expends more energy than other nodes due to its tasks. To save this energy consumption, authors in [54] introduced a new mechanism called V-LEACH (Vice Cluster-Head-LEACH to avoid the

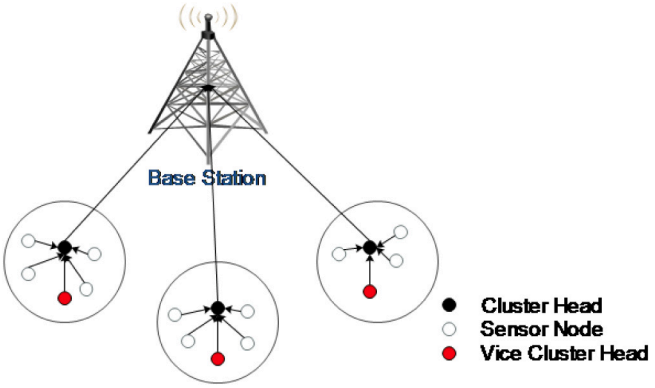


Fig. 7. V-LEACH architecture.

quick death of CHs. This protocol designed a new scheme as shown in Fig. 7.

Once clusters are formed, the V-LEACH algorithm elects a node within the cluster as a vice-CH depending on the remaining energy. In the beginning, normal nodes transfer their collected data to the CH, then this latter to the sink. When the CH dies, the vice-CH takes the role of the CH. This protocol increases the efficiency of the CH election by selecting a node as a vice-CH, which replaces the CH in the event of death. Thus, it assures that data is always transmitted to the sink even if a CH dies in a specific round [55].

An improved centralized algorithm based on LEACH named W-LEACH (Weighted-LEACH) is proposed in [56], where the BS is the unique component that responsible for picking active sensor nodes and sleeping sensor nodes. This centralized data aggregation protocol includes two phases like LEACH: Set-up and steady phase. In the first phase, authors change the **P** definition to become the percentage of the maximum quantity of CHs in place of the current number of CHs as it is set in the classical LEACH protocol.

At the beginning of each round, W-LEACH calculates a weight value **Wi** that it is attributed to every sensor node, without considering if this node was a CH or not in the precedent turns [55]. Residual energy and density of nodes are two principal elements, which calculate the weight of the node (Wi) [13]. Wi value is calculated applying Eq. (11) [56].

$$T(i) = \begin{cases} e_i * d_i, & if \quad d_i > d_{thresh} \\ d_i & \text{otherwise} \end{cases} \tag{11}$$

Where the density of each node is d_i

$$d_i = \frac{1 + number\ of\ alive\ nodes\ in\ range\ r}{n} \tag{12}$$

Where r is the range that is attainable by sensor node i. e_i presents the residual energy of node i, and d_{thresh} is a density threshold to define the set of sensors in very low-density areas [56].

CHs are chosen depend upon the value of Wi, such that nodes with high Wi have a high chance to be selected as CH.

Once CHs are formed, clusters are established such that every sensor node is attributed to its closest CH.

Hnini et al. in [57] improved W-LEACH algorithm by W-LEACH decentralized (WD-LEACH). The WD-LEACH is a decentralized protocol, which serves to prolong the network lifetime in WSNs by using the density of the nodes to split the network into two groups: active nodes and sleeping nodes. Tasks in W-LEACH decentralized is done into epochs in two stages as well as in W-LEACH and LEACH protocols.

Every epoch starts with the set-up stage. In this stage, CHs are picked out and clusters are established. Afterward, the permanent stage is coming, that characterized by data transmission. In this phase, every non-CH node can precise its neighbor and the distance corresponding to every neighbor depending on the density of nodes. Then, it constructs its cluster [58]. Following the number of neighbors, each node can define its status as being an active or stay in a sleep mode over the current turn. Hence, sensor nodes that have low densities remain alive as long as possible [57].

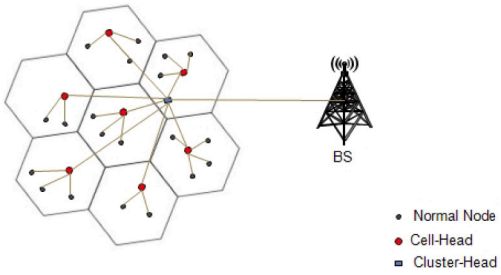


Fig. 8. LEACH-Cell architecture.

W-LEACH decentralized uses the density of sensor nodes to decide the node status either active or sleep status. Thus, in every cluster it allows just a part of nodes to forward data to the CH in the epoch. Consequently, W-LEACH decentralized algorithm can decrease the redundant data CHs and augment the network lifetime. However, it has some disadvantages, such as it can work just only in-network where the collected data is the same.

A WD-LEACH diagram is provided in [57].

Table 3 introduces the advantages and disadvantages of protocols base on a single-hop technique to route data from CHs to the BS.

3.2.2. Multi hop

The multi-hop technique reduces the energy cost used by all nodes along the route. In this category, protocols are based on the multi-hop technique to route data from CHs to the BS.

Among the protocols used to improve network coverage in sensor networks are LEACH-Cell [59], which divides the network into sections named cells. Every seven neighboring cells formed a cluster with a CH. Each cell contains some sensor nodes and a cell-head that is elected within the cell to be a head as we can see in Fig. 8.

The cell-head has the same function as CH in the LEACH approach. That means, it creates a TDMA schedule, allocates a time slot to each cell member, aggregates data from cell members through the TDMA schedule, then forwards the received data to the CH belongs. The CH repeats the same procedure as cell-head in order to gather data from its cell-heads, then it propagates it to the sink through the optimal path. No re-celling and re-clustering are carried out other times because they keep the same in all along the life-cycle of the network once formed, only CHs and cell-heads switch dynamically [59].

The principal purpose of LEACH-Cell is to switch off the nodes transmitter when the cell-head sends data to the CH. Thus, the generation of cell-heads decreases communication cost between nodes and the CH [60].

The pseudo-code for selecting a new cell-head and a new cluster-head is shown in [59].

Owing to the random selection of CHs in the LEACH protocol, some CHs may be gathered on a specific network surface. Consequently, some sensor nodes joined a CH, nonetheless many sensor nodes cannot be connected to any cluster. These sensor nodes are examined as orphan nodes. Hence, orphan nodes' data cannot achieve the BS. To solve this situation, authors in [61] developed O-LEACH (Orphan-LEACH), which allows these nodes to transmit their data to the sink.

O-LEACH addressed two scenarios, the first one considers a cluster member as a gateway, which is capable to take on the role of CH of orphan nodes. Then, a number of orphan nodes are connected to the gateway, and this latter forwards aggregated data to its CH belongs as shown in Fig. 9.

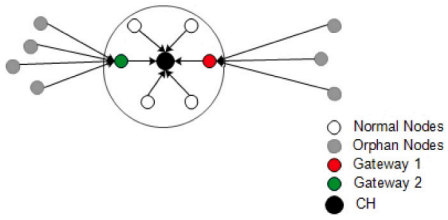


Fig. 9. Scenario 1 general and more precise of orphan nodes in the network.

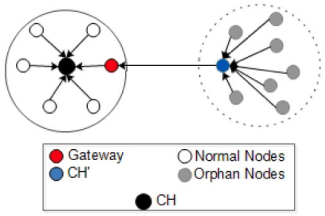


Fig. 10. Scenario 2 of orphan nodes in the network.

The second scenario has been proposed to form a private orphan nodes cluster with a special CH [16]. This scenario is applied when the quantity of orphan nodes overtakes the number of cluster participants of CH. The private cluster consists of a number of orphan nodes and a CH' (special CH). The CH' is the first orphan node that has reached the gateway. Therefore, the CH takes its role as a CH, then it broadcasts information to its cluster members. Afterward, it regroups data, then sends it to the CH through a gateway as shown in Fig. 10. [61].

In summary, in both scenarios, data of orphan nodes reach the BS.

The multi-hop method between nodes and the BS improves the distance of communication. Authors in [62] have proposed the MH-LEACH (Multi-Hop-LEACH) to pick out the optimal target to transfer data from CH toward the BS using intermediate CHs. The principal goal of this protocol is the communication between CHs and using multi-hop to reach the sink.

The MH-LEACH is established in two phases (the initial phase and the steady phase). CHs and clusters are determined in the same way as in LEACH. Afterward, CHs broadcast announcement messages to the network. Each CH creates its routing table according to the received signal power.

After routing table generation, CHs forward their tables to the BS to check whether CH can be in the path of another. Then, the BS returns to pack the checked tables to the CHs. [62] When a data packets did not accept by any CH, the source CH essays another route in its routing table [12]. To simulate the MH-LEACH see algorithm in [63].

The MHT-LEACH (Multi-Hop Technique-LEACH) [64] is a protocol based on the LEACH approach which discusses routing by split the network into two levels depending on the path length between CHs and the BS.

In MHT-LEACH, the CH selection and the cluster formation are similar to the LEACH set-up phase. Instead of forwarding CHs data immediately to the sink [65], MHT-LEACH divided CHs into two groups (internal and external) according to the location betwixt CHs and the sink. If this distance is strictly lower than d_0, this CH belongs to the internal group which communicates directly with the sink. On the other side, if the CH is located equal to or upper d_0, the CH belongs to the external group [66].

Every external CH is able to generate a routing table based on the announcement messages received from the internal CHs [67]. Then, each external CH picks the lowest route of its routing table in order to transmit its data either directly to the BS or to another CH from the internal group. The MHT-LEACH protocol serves to balance energy charge in the network by distributing the set of CHs into two groups, then lead information to the BS through these groups [68]. But, the MHT-LEACH cannot be effective for the large-scale networks [69]. Instructions of the algorithm and a flowchart of the MHT-LEACH are given in [64].

To improve MHT-LEACH, authors in [66] developed a new extension of MHT-LEACH which is referred to as IMHT-LEACH (Improved Multi-Hop Technique-LEACH), that distributes CHs into multi-levels instead of using two levels in MHT-LEACH to classified all CHs in the network [67,70].

Each level has a diameter of $d_0/2$ from the BS. e.g.:

- Level 1 : distance to the BS $(d_{toBS}) < d_0/2$.
- Level 2: $d_0/2 <= (d_{toBS}) < d_0$.
- Level 3: $d_0 <= (d_{toBS}) < 3d_0/2$.
- ... Until level n.

As MHT-LEACH, in IMHT-LEACH, each CH has its own routing table which it transfers its data based on it [71]. Data is routed between CHs from higher levels to lower ones until it achieves the sink [67].

IMHT-LEACH divides the set of CHs into multi-levels in the network depending on the distance from the BS to obtain balanced energy throughout the network and extend the network life cycle [68]. The simulation results show that IMHT-LEACH avoided MHT-LEACH disadvantages. Consequently, the IMHT-LEACH technique is more adapted for a wide network environment [72]. Instructions of the algorithm and a flowchart of the IMHT-LEACH are given in [66].

Alnawafa et al. proposed a dynamic technique named DMHT-LEACH (Dynamic Multi-Hop Technique-LEACH), considering the CH residual energy over the turn [73], to enhance the IMHT-LEACH protocol, which uses a static technique to route data from CHs to the BS. Hence, CH in the DMHT-LEACH can change its data route dynamically many times in the same epoch [74]. The DMHT-LEACH has four stages to do its job: the initial stage, the announcement, the selection, and the routing stage.

The first stage has been done as the set-up stage in the LEACH approach, which serves to elect CHs and establish clusters. Then, every CH generates its own TDMA schedule. This schedule contains allocation for cluster member nodes and allocations for CHS neighboring. Once the first phase was done, CHs are classified into levels according to their distance toward the BS. The wide of each level is $d_0/2$.

The second stage is characterized by an announcement message that it is broadcast by CHs. This message contains information about the CH ID, the CH coordinates, the distance between CH and the BS, then the CH current energy [71].

In the third stage, each CH establishes its routing table based on information collected from lower level CHs in the second stage.

When routing tables are formed, in the routing stage, the DMHT-LEACH algorithm arranges these routing tables and each CH chooses its optimal route to the BS depending on the residual energy of its neighbors and their distances to the BS. Therefore, a cost function is calculated Eq. (13) [71] to organize routes in the routing table.

$$C_{NCH}(i) = C_{NCHP}(i) + \alpha.C_{NCHE}(i) \tag{13}$$

Where

$$C_{NCHP}(i) = \frac{D_i^2}{SCH_{D-BS} - NCH_{iD-BS}} \tag{14}$$

Where, C_{NCHP} represents the distance cost for the NCH (Next CH), C_{NCHE} refers to the energy cost for the NCH. D_i introduces the distance from the SCH (Source CH) to the NCH. SCH_{D-BS} refers to the distance between the SCH and the BS, while NCH_{iD-BS} indicates the distance of the NCH to the sink. The parameter i presents the member in the group of neighboring CHs which are selected by the SCH in this turn. Finally, the α criterion shows the relative weight.

The advantage of the DMHT-LEACH is that it uses a dynamic technique to route data, so the route in the routing table can be changed frequently when the cost function change.

Table 3 introduces the advantages and disadvantages of protocols base on the multi-hop technique to route data from CHs to the BS.

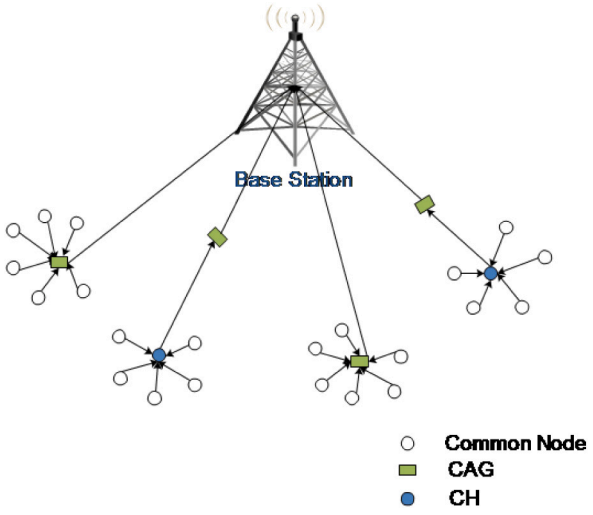


Fig. 11. A-LEACH architecture.

3.2.3. Single and multi hop

In this category, protocols are based on both the single hop and the multi-hop techniques to route data from CHs to the BS.

Advanced-LEACH (A-LEACH) [75] is proposed to solve a drawback of LEACH protocol that the node which is selecting to be CH for the current turn consumes much energy than other normal nodes. A-LEACH is a heterogeneous energy algorithm. Its main aims are prolonging the network lifetime or extending the stability epoch of the network by reducing the risk of nodes failure by using the heterogeneity attribute parameters [75]. A-LEACH executes its tasks in turns like the LEACH algorithm and every turn is split into two stages. A synchronized clock is used to indicate to nodes the beginning of each turn.

In A-LEACH, there are two types of nodes: CAG are nodes that have more energy than others, which will be chosen as CHs or gateways, and the rest of the network nodes represent normal nodes.

In the set-up stage, based on the intended probability of CHs, every node determines its probability of becoming a CH. In the steady stage, when the collected data achieves the CH, this latter regrouped it, then forwarded it to the BS using one of the two ways: either it chooses the nearest gateway (CAG node), which will direct the data to the sink if the CH is normal or it routes the data directly to the sink if it is a CAG node. as we can show in Fig. 11.

The benefits of A-LEACH are:

- It is a decentralized algorithm where clusters are configured independently of the sink.
- It reduced the information quantity that is sensed to the sink through the data fusion.
- Even if all normal nodes die, the CAG nodes keep on forwarding information to the BS.
- The gateway can decrease the energy consumption and the chance of nodes failure, then it prolongs the CHs lifetime.

LEACH-TL [26] solves the unequal energy distribution issue posed by the random selection of CHs in the LEACH protocol. The principal purpose of LEACH-TL is to balance the energy consumption in the network.

Due to the aleatory election of CH, after the first turn, CHs had different remaining energy. So, if a node with low energy or so far from the sink was elected as CH, this latter dies rapidly than other nodes. In this case, LEACH-TL solves this issue by selecting a secondary CH in the cluster that receives data from cluster members, regroups them, then sends them to the primary CH, while the primary CH is responsible to forward data received from the secondary CH to the sink [12]. The LEACH-TL is based on the LEACH approach to elect CHs and form clusters. Afterward, if the remaining energy of the CH is less than the average (see Eq. (15)) [26] or its distance to the sink is more than the average (see Eq. (16) [26]), then this CH requires a secondary CH within its cluster. This latter will be with maximum residual energy. Furthermore, if the CH remaining energy is more than the average, so this cluster does not require a secondary CH [25].

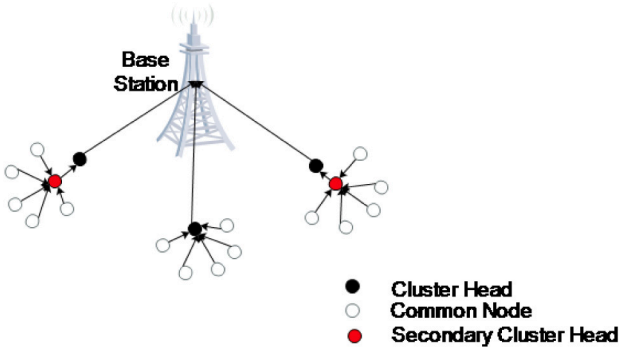


Fig. 12. LEACH-TL architecture.

$$E_{ave} = \sum_{i=1}^{N} E_{cur}(i) \tag{15}$$

$$d_{ave} = \sum_{i=1}^{N} d_i \tag{16}$$

Here, $E_{cur(i)}$ introduces the current energy of the CH, and E_{ave} presents the average energy. The d_i refers to the distance between the CH and the BS, and d_{ave} presents the average distance.

Fig. 12 presents the architecture of the LEACH-TL protocol.

To simulate the LEACH-TL approach, an algorithm is given in [43].

L-LEACH (Energy Balanced) [76] is an energy multi-hop routing protocol, which takes into account only the distance. Unlike LEACH, LEACH-L is adapted for wide range WSNs and the optimal number of the hop is taken.

CHs communicate to the sink in various ways: single-hop and multi-hop according to the distance. They communicate directly to the BS if they are closed to it. On the other hand, when they are further to it they communicate through multi-hop techniques, and the shortest distance of communication is restricted.

The LEACH-L consists of two stages like LEACH. At every turn, a novel CHs are picked out, and the charge is diffused and well-regulated betwixt the set of network nodes. LEACH-L covers up a giant region than Mobile-LEACH due to the fact LEACH-L allocates equal energy amongst all the sensor nodes in the pre-period [31].

Table 3 introduces the advantages and disadvantages of protocols base on single-hop and multi-hop techniques to direct data from CHs to the BS.

The discussed enhanced LEACH hierarchical routing protocols based on data transmission techniques to improve the LEACH algorithm are summarized in Table 4 according to the research domain, network type, modeling parameters of each protocol, and the complexity of the algorithm.

3.3. Protocols based on CH selection and data transmission

In clustering routing protocols in WSNs, both CH selection and data transmission are important and have a significant influence to increase the network life-cycle. Protocols based on CH selection and data transmission consider the multi-hop technique and some specific metrics to select CH, such as residual energy.

Table 3
Advantages and disadvantages of LEACH-based protocols (class 2).

Protocol	Strengths	Limitations
LEACH-F	Offers better CHs distribution using centralization. Reduces the delay of set-up stage.	Clusters are fixed at the beginning. No re-clustering stage. Once clusters are established, adding or taking off nodes is impossible.
V-LEACH	Replaces CH by Vice-CH in case of death of the main CH. Extend the lifespan. There is no require to elect a novel CH each time when CH dies.	The early death of the CH when it is further from the BS. Additional treatment for vice CH.
W-LEACH	Economy energy consumption.	Random CH selection.
WD-LEACH	Dividing the network into active and sleep nodes Allows only a part of sensor nodes to communicate with the CH.	Will not be effective in networks where the data collected is different.
LEACH-Cell	Good coverage of the network. Energy efficiency increase.	Random selection of CHs and cell-heads. Various type of head (CHs and cell-heads) provides overload of control packet.
O-LEACH	Orphan nodes can send their data to the BS. Good coverage of the network.	Random selection of CHs. Communication with the BS in a single hop.
MH-LEACH	A very high energy-efficiency. Use the multi-hop method.	Random CHs election. CHs that group data of other CHs will consume more energy.
MHT-LEACH	A very high energy-efficiency. Using the multi-hop technique between CHs. Classification of CHs into two groups based on the distance to the sink.	Selects CHs without taking into account the energy. The no. of CHs in each turn may be not the same. Only two levels, CHs at the edge of the network consume more energy. The No. of nodes in clusters is not equally.
IMHT-LEACH	Uses the multi-hop technique between CHs. Splits the CHs into multi levels.	Random election of CHs. Considers only the distance to the BS.
DMHT-LEACH	Divides network into multi levels depend upon the distance to the BS. Each CH generates its routing table on basis of the distance and the remaining energy. The route will change once the cost function change.	Random selection of CHs. The no. of CHs in every epoch may be not the same. The no. of nodes in clusters is not equal.
A-LEACH	Extend the stability period. Reduces the risk of nodes failure.	Random CHs election. Based on the single hop for data transmission. Additional treatment of CAG nodes.
LEACH-TL	Energy harvesting. Multiple CHs. Considers the remaining energy and distance for selecting the secondary CH.	Growth in overhead cost. Single hop to send data from CHs to the BS.
L-LEACH	Is adapted for large networks. Uses an optimal number of hop. Regulated network load.	It takes into account only the distance. CHs with less residual energy can die quickly.

Table 4
General characteristics of LEACH-based protocols (Class 2).

Protocol	Year	Research domain	Network type	Modeling Parameters	Complexity
LEACH	2000	Load balancing	homogeneous	Energy saving	Low
A-LEACH	2010	Energy saving	heterogeneous	Energy harvesting ability	Yes
LEACH-Cell	2012	Division of the network	homogeneous	Remaining energy and shortest route	Yes
LEACH-F	2013	Fixing the number of clusters	homogeneous	No. of clusters	Yes
LEACH-TL	2013	Energy saving and Multiple CHs	homogeneous	Remaining energy and distance to the sink	Yes
V-LEACH	2013	Multiple CHs in the cluster	homogeneous	Remaining energy	Yes
O-LEACH	2016	Division of the network	homogeneous	Remaining energy and distance from a orphan node to the gateway	Yes
W-LEACH	2013	Division of the network	homogeneous	Weight value	Yes
WD-LEACH	2014	Division of the network	homogeneous	Density of nodes	Yes
L-LEACH	2013	Network Coverage	homogeneous	Distance	Yes
MH-LEACH	2014	Multi-hop technique	homogeneous	Distance from CHs to the sink	Yes
MHT-LEACH	2016	Division of the network and multi-hop	homogeneous	Distance from CHs to the sink and consumed energy	Yes
IMHT-LEACH	2017	Multi-level and multi-hop	homogeneous	Distance and consumed energy	Yes
DMHT-LEACH	2017	Multi-level and dynamic route	homogeneous	Residual energy and distance	Yes

E-LEACH (Enhanced-LEACH) [77] is an enhanced version based on LEACH. This protocol made changes to the set-up phase and the steady phase.

At the set-up level, the E-LEACH accounts that the remaining energy is the principal metric [53]. Therefore, from the second round, it elects CH according to the residual energy of nodes. That means in each

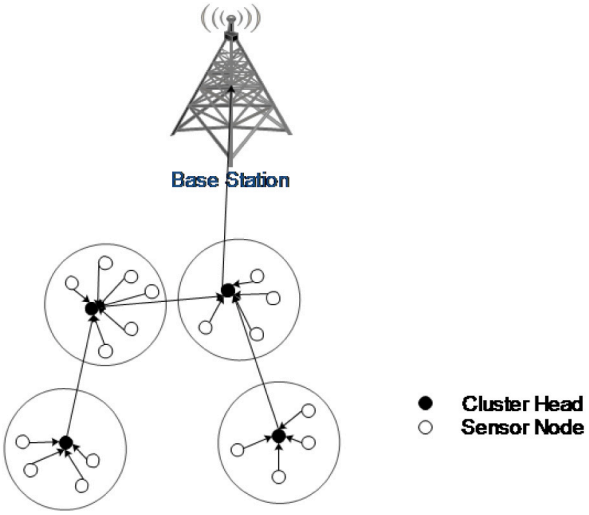


Fig. 13. E-LEACH architecture.

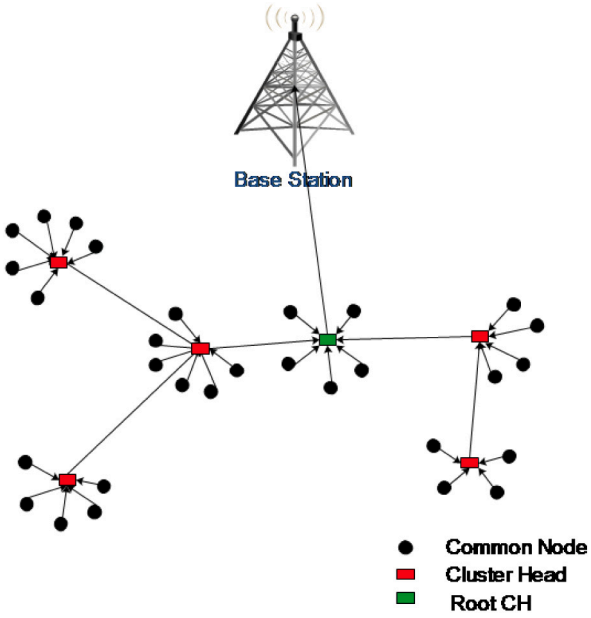


Fig. 14. BRE-LEACH architecture.

round, the node has more remaining energy will be candidate as CH while nodes with few residual energy will be as normal nodes [55]. Authors are defined as a new threshold function [77] as follow:

$$T_{new}(n) = \begin{cases} \frac{P_{head}}{1-P_{head}*(r \bmod (1/P_{head}))} * \frac{E_{curr}}{E_{init}}, n \in G \\ 0 \qquad \text{otherwise} \end{cases} \tag{17}$$

Where

$$P_{head} = \sqrt{\frac{N}{2\pi}} \sqrt{\frac{E_{fs}}{E_{mp}}} \frac{M}{d_{toBS}^2 * N} \tag{18}$$

At the steady level, to improve data communication directly to the BS in a single hop, the E-LEACH protocol proposed to pick out a CH with maximum residual energy as a root-CH which aggregates data from other CHs, then sends them to the BS. Besides, the other CHs use the optimal path betwixt CHs to reach the root.

Fig. 13 presents the architecture of the E-LEACH protocol.

Authors in [25] introduced an enhanced LEACH protocol, namely BRE-LEACH (Balanced Residual Energy-LEACH), which improved the network life cycle, the stability period, and decreased the cost of energy consumption. The CH consumes more energy than other network nodes because it gathers all cluster member's data. Therefore, the random selection of CHs in the LEACH approach without considering the residual energy of nodes causes an energy imbalance, and some nodes can finish their energy quickly than other nodes. So, this problem is solved by the BRE-LEACH algorithm through electing CHs according to the remaining energy and use the threshold introduced in Eq. (19) [25].

$$T_1(n) = \begin{cases} \frac{P}{1-P*(r \bmod (1/P))} * \frac{E_{res}}{E_0}, n \in G \\ 0 \qquad \text{else} \end{cases} \tag{19}$$

Where the chance of the number of CH in the entire network is presented by P, r is the present turn, E_{res} introduces the residual energy of nodes while E_0 defines the beginning energy, then G shows the collection of nodes which have not been CH yet in the last 1/P turns.

As the above-discussed protocols, BRE-LEACH works in turns, and each turn is split into four phases namely: Cluster set-up, TDMA scheduling, Root CH selection, and Data transmission phase.

Once CHs are elected depending on the remaining energy, each sensor node joined its cluster depending on the received signal from CHs. Thus, every CH provides a TDMA schedule for its cluster members. Afterward, the BRE-LEACH algorithm chooses the CH which has residual energy more than the average and the distance to the BS less than the average as a root CH. This root regroups data from other CHs, then forwards it to the sink. On the other hand, the non-root-CH use the multi-hop metric to achieve the parent CH.

Fig. 14 illustrates the architecture of the BRE-LEACH algorithm.

Simulation results in [25] demonstrated that the BRE-LEACH algorithm is better than LEACH in decreasing the cost of energy consumption and prolonging the network lifespan by 55.73% compared to the LEACH algorithm.

To improve the random selection of CH in DMHT-LEACH which influences the network life cycle. Authors in [74] developed an improved version of DMHT-LEACH called EDMHT-LEACH (Enhanced Dynamic Multi-Hop Technique-LEACH), which characterized by a new method to elect CH and forming clusters. Thus, this new approach provides some modifications in the dynamic technique to choose routes. Like IMHT and DMHT, the EDMHT protocol divided the network into multi-levels around the sink. The wide of each level is $d_0/2$. Similar to DMHT, this new version has four stages. The changes made by the EDMHT-LEACH technique are as follows:

- CH selection according to the residual energy of node applying the following equation (20)[74].

$$T(i) = \begin{cases} max(\frac{P}{1-P*(r \bmod (1/P))} * \frac{E_{residual}}{E_{max}}, T_{min}), n \in G \\ 0 \qquad \text{otherwise} \end{cases} \tag{20}$$

- The number of nodes in each cluster (N_0): EDMHT assumes that the number of nodes within the cluster is limited to (N_0) defined in Eq. (21) [74] and cannot exceed it [74].

$$N_0 = \frac{Number\ \ of\ \ sensor\ \ nodes}{Number\ \ of\ \ CHs} \tag{21}$$

- The appearance of IN (Independent Nodes) in the network: these nodes represent nodes that have not joined any CH in the cluster formation phase.

Once CHs are elected, clusters are established and each node recognized its role in the network, CHs and INs create their routing table depending on the announcement messages. Then, routes in routing tables are organized based on the cost function. Hence, as in DMHT, the lowest cost routes will be used to direct CH or IN data to the BS. Then, this route can be changed dynamically if the cost function is changed in the same turn.

Table 5
Advantages and disadvantages of class 3.

Protocol	Strengths	Limitations
E-LEACH	Equilibrating energy consumption in the network. CHs selection depending on the remaining energy of nodes.	Electing CH according to the energy, it may be further from the BS. Chooses a root-CH based only on the energy whatever the distance to the BS.
BRE-LEACH	Extends the network lifetime. CH selection according to the residual energy. Pick out a parent CH depending on the remaining energy and the distance to the BS.	No. of nodes in clusters is not equal. In every turn, the no. of CHs may be is not the same.
EDMHT-LEACH	Considers the remaining energy in CHs selection. Defines the maximum number of nodes in each cluster. Divides network into multi levels. Uses a dynamic route of data.	Uses single hop in intra-clusters.

Table 6
General characteristics of LEACH-based protocols (Class 3).

Protocol	Year	Research domain	Network type	Modeling Parameters	Complexity
LEACH	2000	Load balancing	homogeneous	Energy saving	Low
E-LEACH	2012	Energy-Efficient	homogeneous	Remaining energy	Yes
BRE-LEACH	2019	Energy-Efficiency	homogeneous	Remaining energy and distance to the BS	Yes
EDMHT-LEACH	2017	Multi-level and dynamic route	homogeneous	Residual energy and distance	Yes

The EDMHT-LEACH saves energy by selecting CH according to the residual energy of nodes. Thus, through this technique, also independent nodes can transfer their data to the BS. Therefore, this enhanced approach decreases the total energy consumption and extends the network lifespan. Furthermore, EDMHT-LEACH has some drawbacks such as it uses the same process as LEACH for transmitting data for intra-cluster. So, the dynamic routing technique is used just for inter-cluster [78]. A flowchart of the EDMHT-LEACH protocol is given in [74].

Table 5 introduces the advantages and disadvantages of protocols base on the CH selection and data transmission technique to enhance the original LEACH algorithm.

The discussed improved LEACH hierarchical routing protocols based on the CH selection and data transmission techniques to enhance the LEACH algorithm are summarized in Table 6 according to the research domain, network type, modeling parameters of each protocol, and the complexity of the algorithm.

3.4. Comparative analysis and discussion

Considering the special features of WSN, the clustering technique is the most challenging field in WSN owing to its various advantages, which decrease the number of routed packets and achieve good energy-efficiency in the network.

The basic LEACH approach faces numerous restrictions in terms of CH selection and data transmission techniques, which impact the network lifetime and increase energy consumption.

This survey classified LEACH variants clustering protocols into CH selection-based routing protocols, data transmission based routing protocol, and protocols based on both CH selection and data transmission to enhance the LEACH performances.

These protocols based on LEACH were proposed to improve the classical LEACH approach through decreasing energy consumption and extending the lifetime of WSNs.

The discussed LEACH-descendants exhibit better performance than the original LEACH protocol by considering the CH selection approach and data transmission techniques.

Table 7 provides the improvements made by LEACH derived protocols compared to classical LEACH.

A comparative analysis of LEACH variants has been highlighted as shown in Tables 8 and 9.

The discussed clustering routing protocols are compared based on different aspects, such as clustering method (centralized or distributed), CH selection method (either random or considering some parameters like residual energy, distance, the density of node, etc.), mobility of node, scalability, deployment of nodes, communication techniques (single-hop or multi-hop or both), energy-efficiency, localization information.

The set of the discussed protocols use the TDMA schedule which allows nodes to be in standby mode while waiting for their data transmission turn. They consider that the sink is fixed in a defined place, and then they are based on the data aggregation technique. Finally, all the discussed protocols use random rotation.

The random selection of CH in the network affects the entire performance of WSN. Due to the tasks carried out by the CH in collecting, aggregating data, and sending them to the sink, the CH selection impacts energy consumption and the life-cycle of WSN. So, the right choice of CH node is a challenge in routing protocols. Several LEACH-based clustering protocols were developed to deal with this issue based on different aspects. As illustrate in Table 9, numerous of discussed CH selection-based protocols pick out the CH by considering the residual energy parameter, such as the LEACH-B and the I-LEACH. LEACH-C uses the centralization technique at the BS and the residual energy to select CH and form clusters in the network. Furthermore, LEACH-M and LEACH-ME are based on the mobility of nodes and residual energy. VH-LEACH, LEACH-T, and TB-LEACH utilize the energy efficiency parameter. Then, Improved-LEACH, EADCR, and BN-LEACH consider distance and residual energy.

Another essential problem in designing routing protocols in WSN is data transmission techniques, which have an important impact on the network lifetime. In the single-hop transmission, the CH sends the aggregated data from its cluster members directly to the BS as used in the basic LEACH protocol. This technique is useful only in small network areas.

On the other hand, the multi-hop technique is used to enhance data transmission. In the multi-hop communication, the CH uses some intermediate nodes (either nodes or other CHs) to forward its data to the BS. The principal goal of the multi-hop technique is to maintain the distance between the source and the destination node less than the threshold distance; because the energy dissipation in the radio model is proportional to the distance. If the distance between the source and the destination nodes is less than the threshold, the energy dissipation is in the function of d^2. On the other side, the energy consumption is in the function of d^4.

Table 7
Comparison of LEACH and its descendants.

Protocol	Difference from the original LEACH
A-LEACH	It aims to improve the stability, energy-saving, and reduce the chance of nodes failure by using CAG nodes.
LEACH-B	In each turn, LEACH-B guarantees a near-optimal number of CHs. Then it takes into account the residual energy for selecting CHs after the first turn.
LEACH-C	Centralized algorithm, the BS creates clusters and elects CHs according to nodes location and the residual energy of nodes.
LEACH-Cell	LEACH-Cell splits the network into cells (sections), and every seven close cells form clusters. Every cell contains a cell-head, and each cluster has a CH.
E-LEACH	CH selection based on the remaining energy of nodes, and an optimal CH (with maximum remaining energy) is elected to group other CHs packets, then direct data directly to the sink
LEACH-F	CHs selection and clusters formation are based on the centralizing approach. Thus, clusters are fixed once they are formed at the beginning.
LEACH-M	LEACH-M uses mobility of both normal nodes and CHs for transfer data to the BS. Therefore, it is adapted for mobility environments.
LEACH-ME	Sensor nodes with minimum mobility compared to its neighbors are selected as CHs.
I-LEACH	CHs Election on the basis of the remaining energy, nodes location, and neighbor number. Then, it updates the threshold function by adding the residual energy. Thus, I-LEACH envisages the optimal number of clusters.
LEACH-TL	In LEACH-TL, two kinds of CHs are generated: primary and secondary CH. When the primary CH has remaining energy less than the average or its distance to BS is more than the average, The secondary CH takes the role of CH and sends aggregated data to the primary CH, whereas this latter communicates only with the BS.
V-LEACH	When a Cluster-Head dies, a vice Cluster-Head replaces it.
VH-LEACH	CHs selection on the basis of the remaining energy, and when a cluster head's residual energy attains a threshold, a VH node takes the role of the CH.
O-LEACH	Orphan nodes can join clusters and send their data to the sink.
LEACH-T	Fixation of the number of time slots in each CH TDMA. LEACH-T considers the residual energy for establishing clusters, and it envisages an optimal number of CHs in the network.
TB-LEACH	The number of CHs in every epoch is fixed. Thus, CHs selection in TB-LEACH depends on a random timer.
Improved-LEACH	The distance and the residual energy are two factors that are considered in selecting CHs.
EADCR	Improve the network lifetime by applying the FCM approach to establish clusters, selecting CHs using the fitness function which depends upon the Euclidean distance and the residual energy of nodes.
BN-LEACH	Using the Bayesian Network model to pick out CHs.
W-LEACH	In W-LEACH, there are two types of nodes: active and sleeping nodes, that they are attributed by the base station. Thereby, CHs are chosen depending on the weight value Wi that it is computed based on the residual energy and the density of nodes.
WD-LEACH	WD-LEACH divides the network into active and sleeping nodes without using centralization in the BS using the number of neighbor nodes.
L-LEACH	L-LEACH is based on the distance to choose CHs. The multi-hop method is used as the single hop depending on the distance from the base station.
BRE-LEACH	Only nodes with enough remaining energy can participate in CH selection. BRE-LEACH updates the threshold formula by adding the ratio of residual energy per initial energy. Thus it elects the root CH with maximum energy and minimum distance to the BS to group data from CHs, and sends it to the sink.
MH-LEACH	In place of forwarding packets to the sink directly, MH-LEACH uses the multi-hop technique between CHs to achieve the sink through constructing a routing table for each CH.
MHT-LEACH	MHT-LEACH divides CHs into two groups according to the distance from the BS. Thus, it uses multi-hop between CHs to reach the BS.
IMHT-LEACH	IMHT-LEACH divides CHs into multi-levels, and each level has a length of $d_0/2$. Each CH sends its data to the BS using the multi-hop technique according to its own routing table.
DMHT-LEACH	CHs are distributed into multi-level, and each level has a length of $d_0/2$. Each CH creates its routing table according to a cost function. Thus, CH can change its data route dynamically several times in the same turn once the cost function change .
EDMHT-LEACH	EDMHT-LEACH updates the threshold function to choose CHs to depend upon the remaining energy of nodes. It determines the maximum number of nodes that it can be in a cluster. CHs and INs nodes (Independent nodes) generate their routing table according to a cost function. The route in routing tales are organized dynamically on the basis of the cost function.

The use of single-hop or multi-hop is depending on the application and the network (small or large). For example, if the network is small, it is not worth using the multi-hop because the single-hop is enough and the multi-hop will consume more energy. On the other hand, if the network is wide, the multi-hop is necessary to minimize the energy consumption because the single-hop requires more energy which influences the lifetime of the entire network.

This survey provides a comprehensive study of LEACH descendant clustering protocols, which classify them into protocols that improve LEACH by improving the CH selection phase, protocols that enhancing LEACH through enhancing data communication (single-hop and multi-hop techniques), and protocols that evolve both CH selection and data communication techniques.

Our survey grouped the improved LEACH protocols based on data communication into three groups: (1) single-hop-based protocols, such as LEACH-F, V-LEACH, W-LEACH, and WD-LEACH; (2) multi-hop-based protocols to enhance LEACH performances, such as LEACH-cell, O-LEACH, MH-LEACH, MHT-LEACH, IMHT-LEACH, and DMHT-LEACH; and (3) protocols that used both routing techniques to improve the classical clustering LEACH protocol, such as A-LEACH, LEACH-TL, and L-LEACH.

Finally, in the discussed protocols, some protocols improve both the CH selection by using the residual energy parameter and use the multi-hop to enhance data transmissions, such as the E-LEACH, BRE-LEACH, and the EDMHT-LEACH.

From this survey and the comparative analysis, some conclusions will be concluded: designing a good routing protocol faces to two major constraints, selecting a reliable and an efficient CH is necessary, also using an efficient data communication techniques is important to extend the network lifetime and reducing the energy consumption.

Table 10 shows the improvement of the network lifetime and the energy consumption of LEACH-variant protocols compared to LEACH.

Some numerical results of LEACH-descendant protocols are given in the literature, such as in I-LEACH, VH-LEACH, TB-LEACH, Improved-LEACH, BN-LEACH, V-LEAH, W-LEACH, MH-LEACH, L-LEACH, E-LEACH, and BRE-LEACH. Thereby, through numerical results given

Table 8
Characteristics of LEACH-based routing protocols.

Class	Category	Protocol	No. of CHs	No. of nodes in cluster	Clustering method	Localization
CH selection	Residual energy	LEACH	Uncertain	Uncertain	Distributed	No
		LEACH-B	Certain	Uncertain	Distributed	Yes
		I-LEACH	Certain	Uncertain	Distributed	No
	Centralization	LEACH-C	Certain	Uncertain	Centralized	Yes
	Mobility	LEACH-M	Uncertain	Uncertain	Distributed	Yes
		LEACH-ME	Uncertain	Uncertain	Distributed	Yes
	Energy efficiency	VH-LEACH	Uncertain	Uncertain	Distributed	Yes
		LEACH-T	Certain	Uncertain	Distributed	No
		TB-LEACH	Certain	Uncertain	Distributed	No
Data transmission	Distance	Improved-LEACH	Uncertain	Uncertain	Distributed	Yes
		EADCR	Certain	Uncertain	Centralized	Yes
		BN-LEACH	Uncertain	Uncertain	Distributed	Yes
	Single Hop	LEACH-F	Certain	Uncertain	Centralized	Yes
		V-LEACH	Uncertain	Uncertain	Distributed	Yes
		W-LEACH	Uncertain	Uncertain	Centralized	Yes
		WD-LEACH	Uncertain	Uncertain	Distributed	Yes
	Multi Hop	LEACH-Cell	Uncertain	Uncertain	Distributed	Yes
		O-LEACH	Uncertain	Uncertain	Distributed	No
		MH-LEACH	Uncertain	Uncertain	Distributed	Yes
		MHT-LEACH	Uncertain	Uncertain	Distributed	Yes
		IMHT-LEACH	Uncertain	Uncertain	Distributed	Yes
		DMHT-LEACH	Uncertain	Uncertain	Distributed	Yes
	Single/Multi Hop	A-LEACH	Uncertain	Uncertain	Distributed	No
		LEACH-TL	Uncertain	Uncertain	Distributed	Yes
		L-LEACH	Uncertain	Uncertain	Distributed	Yes
CH selection and Data transmission		E-LEACH	Certain	Uncertain	Distributed	No
		BRE-LEACH	Uncertain	Uncertain	Distributed	Yes
		EDMHT-LEACH	Uncertain	Certain	Distributed	Yes

in the literature, we have calculated the percentage of the network lifetime of some protocols, such as LEACH-B, VH-LEACH, LEACH-T, LEACH-TL, MHT-LEACH, IMHT-LEACH, and EDMHT-LEACH. On the other hand, due to a lack of numbers, we cannot compute the percentage of the improvement of network lifetime and the energy consumption of LEACH-C, LEACH-M, LEACH-ME, EADCR, LEACH-F, WD-LEACH, LEACH-cell, O-LEACH, DMHT-LEACH, and A-LEACH.

Another comparison between protocols is given as follows:

- In [34], the network lifetime is increased with I-LEACH by 59% than the DBS approach, and as well as 29% than LEACH-C protocol. Thus, the energy consumption in I-LEACH has been decreased by 55% than DSB, and by 27% than the LEACH-C algorithm. In addition to the number of data packets that are forwarded to the BS has been raised to 56% using the I-LEACH approach in comparison to LEACH, 21% than LEACH-C, and 46% than DSB.
- LEACH-C forwards about 40% more data per unit energy than LEACH [36].
- In the LEACH-ME, the successful communication rate is 16% at the mobility factor of 4.0, which is better than the LEACH-M. For high mobility (mobility factor 4 and above), the overhead of LEACH-ME is 22% more than that of the overhead of LEACH-M [41].
- The LEACH-VH protocol consumes 26.69% of the total energy, which is about 41.178%, 27.31%, 18.418%, 17.088%, 13.596%, and 11.578% less than LEACH, LEACH-C, LEACH-F, LEACH-L, TL-LEACH, and V-LEACH, respectively [44].
- Some percentage comparison between the EADCR and other protocols like FCM, REHR is provided in [49].
- In terms of the delay of the FND (first node death), the BN-LEACH performs better by 19.5%, 46.06%, and 52.61% than LEACH-C, WEEC, and LEACH, respectively. In terms of the delay of the LND (last node death), the BN-LEACH performs better by 23.54%, 28.32%, and 43.55% than WEEC, LEACH-C, and LEACH.
- In [62], there are two scenarios, in the 1st scenario (50 nodes, area of 50X50 m^2, and BS(25,100)), the MH-LEACH decreases energy consumption by 9%, and increases the network lifetime by 4.57% compared to LEACH. In the 2nd scenario (100 nodes, area of 70X70 m^2, and BS(35,140)), the MH-LEACH reduces energy consumption by 6% and enhances the network lifetime by 4.61% compared to LEACH.

A complete comparison among the current survey and previous surveys in [12–16] is given in Table 11.

4. Future works and research challenges

The comparative analysis and discussions exhibit that both CH selection and data transmission techniques are the major challenges in designing routing protocols in WSN. All the discussed clustering protocols associated with LEACH focused on reducing energy consumption and maximizing the network lifetime. For this reason, various approaches have been proposed to improve the basic LEACH protocol. A variety of parameters have been considered by these approaches. But, there is nonetheless a wide variety of open research problems to be addressed in the future.

The main objectives for developing LEACH-derived protocols for WSNs are: enhancing the WSN performance, energy-efficiency, energy distribution between nodes, enhancing the scalability, selecting the best node to be the CH in each cluster, increasing the security in WSN, reduction of network delay, increasing the stability of the network.

This survey proposes some future issues as follows:

Table 9
Comparison based on different metrics.

Class	Category	Protocol	CH selection	Communication method	Mobility	Scalability	Energy Efficiency	Residual energy	Deployment
CH selection		LEACH	Random	Single Hop	Static	Limited	High	No	Random
	Residual energy	LEACH-B	Random, Residual Energy	Single Hop	Static	Good	Very High	Yes	Random
		I-LEACH	Residual Energy	Single Hop	Static	Very Good	Very High	Yes	Random
	Centralization	LEACH-C	Residual Energy	Single Hop	Static	Good	Very High	Yes	Random
	Mobility	LEACH-M	Residual Energy, Mobility	Single Hop	Mobile	Very Good	Very High	Yes	Random
		LEACH-ME	Residual Energy, Mobility	Single Hop	Mobile	Limited	High	Yes	Random
	Energy efficiency	VH-LEACH	Residual energy	Single Hop	Static	Very Good	Very High	Yes	Random
		LEACH-T	Residual Energy	Single Hop	Static	Good	High	Yes	Random
		TB-LEACH	Random	Single Hop	Static	Limited	High	No	Random
	Distance	Improved-LEACH	Distance, Residual Energy	Single Hop	Static	Good	High	Yes	Random
		EADCR	Euclidean Distance, Residual Energy	Single hop/Multi-hop	Static	Good	Very High	Yes	Random
		BN-LEACH	Density, distance residual energy	Single hop	Static	Good	High	Yes	Random
Data transmission	Single Hop	LEACH-F	Random	Single Hop	Static	Limited	Very High	No	Random
		V-LEACH	Random	Single Hop	Static	Good	Very High	No	Random
		W-LEACH	Random	Single Hop	Static	Good	High	No	Random
		WD-LEACH	Random	Single Hop	Static	Good	High	No	Random
	Multi Hop	LEACH-Cell	Random	Multi-Hop	Static	Very Good	Very High	No	Random
		O-LEACH	Random	Multi-Hop	Static	Good	High	No	Random
		MH-LEACH	Random	Multi-Hop	Static	Good	Very High	No	Random
		MHT-LEACH	Random	Multi-Hop	Static	Good	Very High	No	Random
		IMHT-LEACH	Random	Multi-Hop	Static	Good	Very High	No	Random
		DMHT-LEACH	Random	Multi-Hop	Static	Good	Very High	No	Random
	Single/Multi Hop	A-LEACH	Random		Static	Good	Very High	No	Random
		LEACH-TL	Random	Single Hop/Multi-Hop	Static	Very Good	Very High	No	Random
		L-LEACH	Random		Static	Very Good	Very High	No	Random
CH Selection and Data transmission		E-LEACH	Residual Energy	Multi-Hop	Static	Very Good	Very High	Yes	Random
		BRE-LEACH	Residual energy	Multi-Hop	Static	Very Good	Very High	Yes	Random
		EDMHT-LEACH	Residual energy	Multi-Hop	Static	Good	Very High	Yes	Random

- Complexity: Owing to the composition of the sensor node, which is characterized by a restricted calculation capability, and small memory. So, the complexity influences the global protocol performance in WSNs. Consequently, scheming a clustering routing protocol with low complexity is very crucial. It must that the clustering routing protocol takes into consideration many parameters with the lowest complexity to minimize the energy consumption in the network.
- Deployment: Deployment is one of the major problems in WSN due to it presents an important parameter to determine the network performance. Some routing protocols improve the LEACH algorithm without considering the deployment of nodes in the network. In some cases, the deployment of nodes increases the energy consumption in the entire network. Therefore, the deployment of nodes in the network must be made by considering some metrics, such as the coverage, energy, connectivity, and so forth.
- Fault tolerance: Fault tolerance is another important problem in WSN because it affects the network lifetime. In WSN, nodes can be deployed in a surrounding environment that cannot be accessed by humans and keeps unattended. So the component failure of a node is inevitable. Especially, if the CH breaks down, it results in more damage to the overall network because the CH is responsible to route its cluster members data to the BS. Consequently, implementing a fault-tolerance parameter in clustering routing protocols is important to avoid any fault caused by link breaks, death of a node, and especially how to re-group and select another CH in case of CH failure.
- Modulation techniques: The choice of modulation and channel encoding with optimal parameters help to reduce energy consumption and ensure greater safety in WSNs. In this context, an optimal modulation with an adapted channel coding method is an extremely crucial technique in the physical layer in WSN. So, implementation of modulation and coding techniques in LEACH-descendant protocols are future opportunities for researchers.
- QoS: Different clustering protocols of WSN have been considered energy-efficiency as a significant key to improving the network scalability and the network lifetime. Nonetheless, some recent applications that use the WSNs have diverse constraints, which require QoS (Quality of Service) metrics in terms of end to end delay, reliability, bandwidth, scalability, latency, adaptivity, throughput, robustness, etc [79]. Therefore, researchers have the opportunity to propose and develop clustering-based energy efficiency protocols based on the QoS (reliability, bandwidth, scalability, latency, adaptivity, throughput, robustness, and so forth) to address efficiency issues in WSNs.
- Meta-heuristic: Meta-heuristic (or heuristic) are nature-inspired algorithms that have been used to develop solutions to deterministic combinatorial optimization issues in WSN. These approaches have been proposed to solve clustering problems, such as the selection of CH, the optimal amount of CH in the network, deployment of nodes, data aggregation, and localization [80]. Meta-heuristic algorithms are important open research challenges for improving LEACH-descendant protocols. There are several improvements that can be adopted in the meta-heuristic algorithms in order to implement them in the LEACH variant protocols to enhance the network lifetime. Future researchers may develop hybrid protocols characterized by the benefits of several meta-heuristics. These hybrid protocols take into account the majority of QoS, adaptive modulation coding techniques, etc to improve WSN performance.
- Security: Encryption security in clustering protocols based on LEACH present a major constraint in WSN due to the nature of the sensor node, which has a small memory and a restricted calculation capability. Owing to the important tasks performed by the CH such as aggregating data and transmitting it to the BS, it represents the device most accessible by hackers. Therefore, authentication and encryption algorithms should be applied at the CH level to prevent attacks.

Table 10
Improvement of the network lifetime and energy consumption of protocols compared to LEACH.

Protocol	Network lifetime	Energy consumption
LEACH-B	25%	–
I-LEACH	67%	62%
VH-LEACH	59.04%	41.178%
LEACH-T	60%	–
TB-LEACH	20 to 30%	–
Improved-LEACH	25%	–
BN-LEACH	52.61%	–
V-LEACH	49.37%	–
W-LEACH	60%	–
MH-LEACH	4.57%	9%
MHT-LEACH	77.70%	–
IMHT-LEACH	55.31%	–
LEACH-TL	52.76%	–
L-LEACH	51%	–
E-LEACH	49%	–
BRE-LEACH	55.73%	55.73%
EDMHT-LEACH	88.43%	–

In future work, we will propose a clustering routing protocol that will address as many issues as possible. It will take into account several parameters of the QoS, more security, tolerates failures, and especially in the case of death CH with the least complexity. It will also take into account the deployment of nodes and will not be deployed randomly as the protocols discussed in this survey. Finally, our future work will take into account modulation and coding techniques to improve the efficiency, reliability, and lifetime of the network.

5. Conclusion

Numerous routing protocols based on the clustering technique have been suggested owing to its advantages. These advantages like data aggregation, load balancing, less energy consumption, and scalability. LEACH is treated as the first classical energy-efficient protocol using the clustering routing. It was proposed to raise the lifetime and decrease the consumption of energy in the network. Nevertheless, LEACH has several drawbacks in connecting with energy efficiency, CH selection, and data transmission, which limited the network life span. To surmount these issues, many routing protocols have been suggested by the researcher to enhance the conventional LEACH protocol. In this survey, the original LEACH approach has been discussed with its benefits and drawbacks. Furthermore, based on the drawbacks of LEACH, we have surveyed 27 LEACH descendants routing protocols. These discussed LEACH-based protocols are given in a new taxonomy, which depends on the CH selection methods and the data transmission techniques. The new taxonomy classified these protocols into three classes: (1) protocols that improve LEACH by improving the CH selection phase; (2) protocols that enhancing LEACH through enhancing the data transmission techniques, and protocols that evolve both CH selection and data communication techniques. Besides, to make this classification more sunny and clear, we divided each class into categories, such as residual energy, energy efficiency, single-hop, multi-hop, and so forth. These metrics are used to organize the discussed protocols owing to their impact on the stability and the lifetime of the WSN. This research discussed 27 clustering algorithms with the LEACH approach. The benefits and drawbacks of each discussed protocol are given in this work, also the difference from the original LEACH. Furthermore, to evaluate and compare these LEACH descendants protocols, this survey examined many metrics, such as the clustering method, self-organization, the number of CHs, the number of nodes in clusters, energy-efficiency, scalability, the communication technique, and so forth. The objective of this survey is to classify the improved LEACH protocols to facilitate the choice of efficient protocols for researchers according to their needs.

Table 11
Comparison among this survey and the existing surveys.

	Current Survey	[12]	[13]	[14]	[15]	[16]
Classification of protocols	Yes	No	No	No	No	Yes
Well detailed taxonomy	Yes	No	No	No	No	No
Detailed description of distinct protocols	Yes	No	No	No	No	No
Survey range	CH selection and data transmission protocols	Global presentation of LEACH and other hierarchical routing protocols	CHs selection protocols	Communication techniques protocols	Overview of LEACH and its improved protocols	Election of CHs and cluster formation routing protocols
No. of parameters used to compare protocols	17	None	5	None	4	6
Advantages and disadvantages of each protocol	Yes	No	No	Yes	No	Yes
Comparison with preceding reviews	Yes	No	No	No	No	Yes
No. of discussed protocols	27	6	12	10	12	21

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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